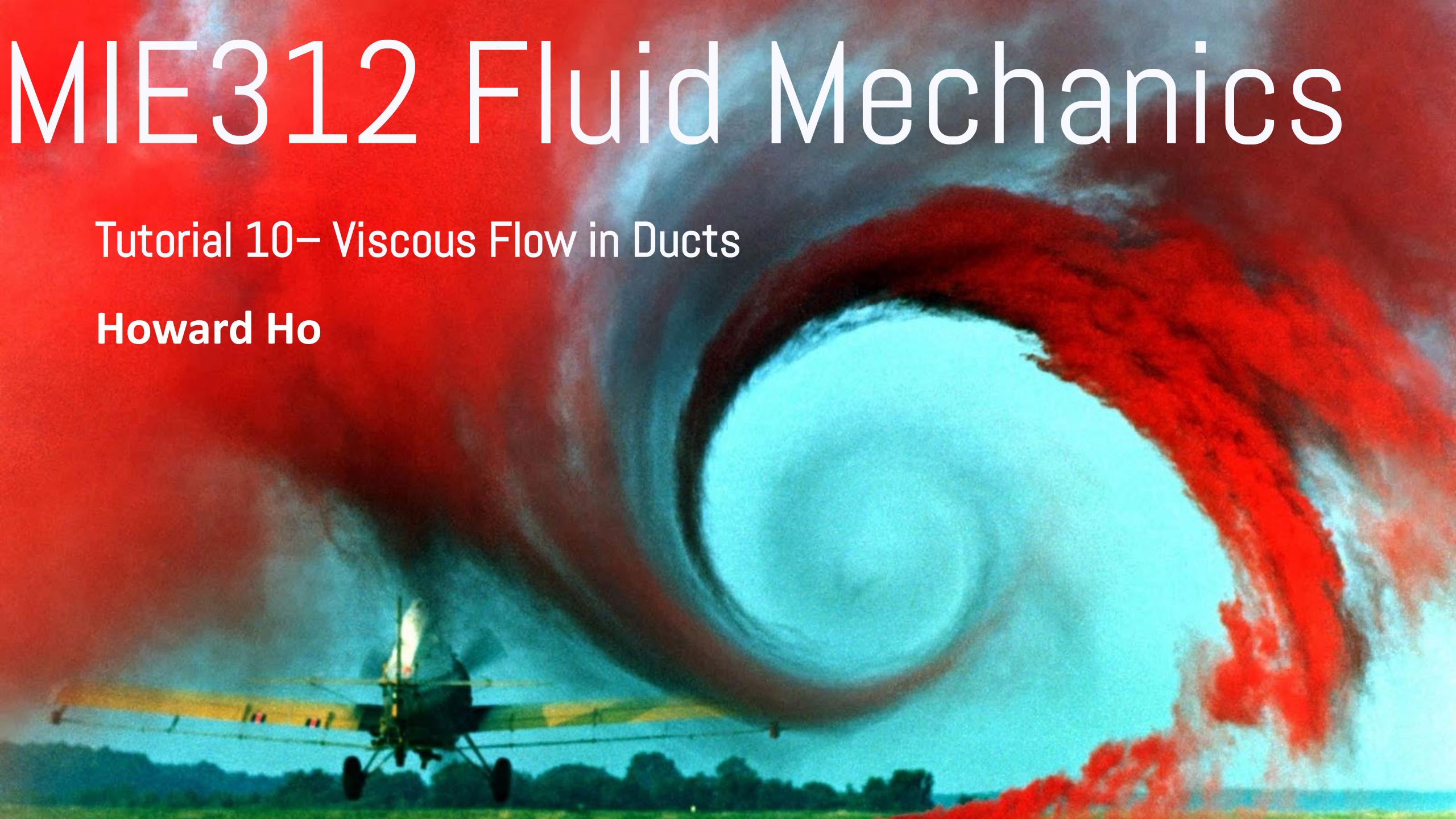
MIE312 Fluid Mechanics

Tutorial 10– Viscous Flow in Ducts

Howard Ho



Learning Objective

- Learn to differentiate Major and Minor loss
- Locate Friction Factor with Moody Chart
- Calculating Pipe Friction Loss and apply with Energy Transport (RTT)





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Recap: Energy Equation

- The energy transport equation allows us to look at pump and turbine problems, note it is no longer the Bernoulli Equation (even if they look similar)

$$\left(\frac{p}{\gamma}+\frac{V^2}{2g}+z\right)_{in}=\left(\frac{p}{\gamma}+\frac{V^2}{2g}+z\right)_{out}+h_{friction}-h_{pump}+h_{turbine}$$

Recall: $\gamma = \rho g$



Friction Head Loss

A big part of chapter 6 is focused around calculating this term

Pump Head Input

Turbine Head Extraction

To find Power: $P = \gamma Q\ h$

$$P_{turbine} = \gamma Q\ h_{turbine}$$

Standard Bernoulli Equation:

$$P_1+\frac{1}{2}\rho V_1^2+\rho g z_1 = P_2+\frac{1}{2}\rho V_2^2+\rho g z_2$$

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Major Losses (Friction)



- Major losses are due to pipe friction, it is calculated as:

$$h_f = f(\frac{L}{D})(\frac{V^2}{2g})$$

where:

- f is the friction factor
- V is the average velocity
- D is the pipe diameter
- L is overall pipe length (this can sometimes be in different unit than pipe diameter)

Note: h_f and major loss are the same thing



The Friction factor f can be found with **Moody Chart** and the **Haaland equation**:

[Use both methods]

$$\frac{1}{\sqrt{f}} \approx -1.8 \log\left(\frac{6.9}{Re_d} + \left(\frac{\varepsilon/d}{3.7}\right)^{1.11}\right)$$

Not this Haaland. It was a Professor in Norway that published this in 1983, the error is only around 2% max



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Minor Losses (Fittings)

- Minor losses arise from fittings such as bends, valves, and expansions. The head loss from these fittings is calculated as:

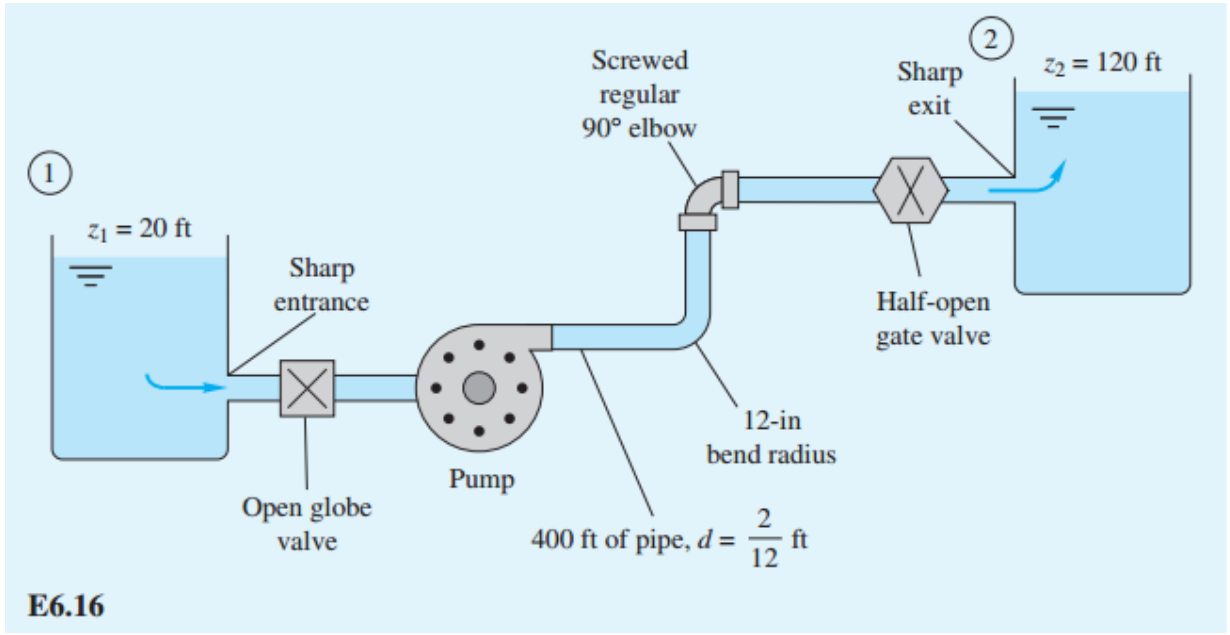
$$h_m = K_L\left(\frac{V^2}{2g}\right)$$

where:

- K_L is the combined loss coefficient of all fittings
- Some useful K to have:
  - Sharp Entrance $K \approx 0.5$
  - Sharp Exit $K \approx 1.0$

Table 6.5 Resistance Coefficients $K = h_m/[V^2/(2g)]$ for Open Valves, Elbows, and Tees

	Nominal diameter, in								
	Screwed				Flanged				
	$\frac{1}{2}$	1	2	4	1	2	4	8	20
Valves (fully open):									
Globe	14	8.2	6.9	5.7	13	8.5	6.0	5.8	5.5
Gate	0.30	0.24	0.16	0.11	0.80	0.35	0.16	0.07	0.03
Swing check	5.1	2.9	2.1	2.0	2.0	2.0	2.0	2.0	2.0
Angle	9.0	4.7	2.0	1.0	4.5	2.4	2.0	2.0	2.0
Elbows:									
45° regular	0.39	0.32	0.30	0.29					
45° long radius					0.21	0.20	0.19	0.16	0.14
90° regular	2.0	1.5	0.95	0.64	0.50	0.39	0.30	0.26	0.21
90° long radius	1.0	0.72	0.41	0.23	0.40	0.30	0.19	0.15	0.10
180° regular	2.0	1.5	0.95	0.64	0.41	0.35	0.30	0.25	0.20
180° long radius					0.40	0.30	0.21	0.15	0.10
Tees:									
Line flow	0.90	0.90	0.90	0.90	0.24	0.19	0.14	0.10	0.07
Branch flow	2.4	1.8	1.4	1.1	1.0	0.80	0.64	0.58	0.41



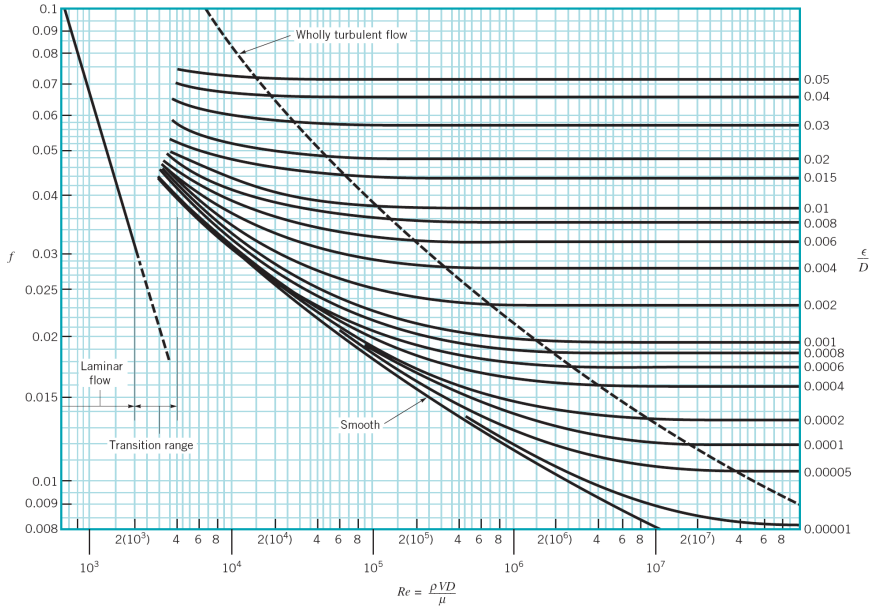
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Steps To Solve Pipe Problems

1. Calculate Velocity (V), often only flow rate is provided
2. Calculate Reynolds Number, $Re = \frac{\rho VD}{\mu} = \frac{VD}{\nu}$
3. Find Friction Factor (f)
   a) Moody Chart first
   b) Then use Haaland equation to validate: $\frac{1}{\sqrt{f}} \approx -1.8 \log\left(\frac{6.9}{Re_d} + \left(\frac{\varepsilon/d}{3.7}\right)^{1.11}\right)$
4. Calculate Major Loss, $h_f = f(\frac{L}{D})(\frac{V^2}{2g})$
5. Calculate Minor Loss, $h_m = K_L(\frac{V^2}{2g})$
6. Combine everything into the Energy transport equation

$$\frac{P_1}{\gamma} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + z_2 + h_f + h_m + h_{\text{turbine}} - h_{\text{pump}}$$



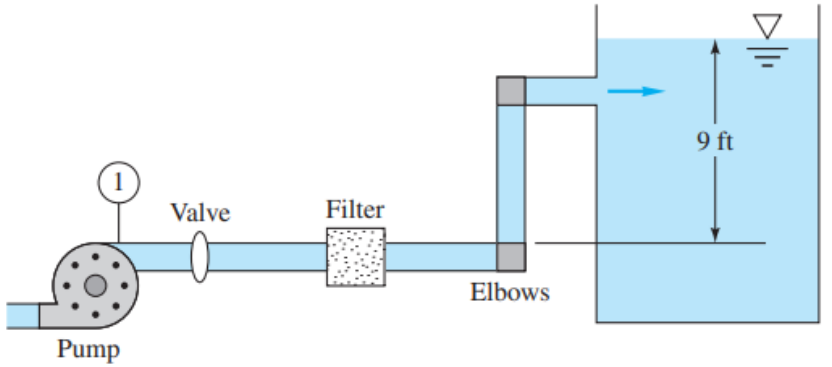
6 Steps Guide
1) Find V
2) Find Re
3) Find f
   a) Moody
   b) Equation
4) Find h_f
5) Find h_m
6) RTT (Energy)

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Problem 1

The water pump maintains a pressure of 6.5 psig at point 1. There is a filter, a half-open disk valve, and two regular screwed elbows. There are 80 ft of 4-inch diameter commercial steel pipe. (a) If the flow rate is $0.4\ ft^2/s$, what is the loss coefficient of the filter? (b) If the valve is wide open and $K_{filter} = 7$, what is the resulting flow rate.

Given: $\rho = 1.94$ slug/ft^3, $\mu = 2.09$ e-5 slug/ft.s, $\epsilon = 0.00015$ ft



7. MINOR LOSSES (K_L) & PIPE ROUGHNESS (ϵ)

Ent. (Sharp)	0.5	Elbow (90°)	0.9
Ent. (Rnd)	0.04	Gate (Open)	0.2
Exit (All)	1.0	Globe (Open)	10
Check Valve	2.0	Tee (Branch)	1.8

ϵ (mm)

Drawn tubing, PVC	0.0015	Cast iron	0.26
Commercial steel	0.046	Riveted steel	0.9–9
Galvanized iron	0.15	Concrete	0.3–3

6 Steps Guide

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3. CONSERVATION LAWS (CV)

RTT: $\frac{dB_{sys}}{dt} = \frac{\partial}{\partial t}\int_{CV}\rho\beta d\forall + \int_{CS}\rho\beta(\mathbf{V}_r\cdot\mathbf{n})dA$

Relative Vel: $\mathbf{V}_r = \mathbf{V}_f - \mathbf{V}_{cv}; \quad \dot{m}_{rel} = \rho A V_r$

Rotating outlet: $\mathbf{V} = \mathbf{W} + \mathbf{U}; \ \mathbf{U} = \boldsymbol{\omega}\times\mathbf{r}$

Steady 1-D Ang. Mom.: $M_z = \sum_{out}\dot{m}\,rV_\theta - \sum_{in}\dot{m}\,rV_\theta$

Mass: $\sum\dot{m}_{in} = \sum\dot{m}_{out} + \frac{\partial m_{cv}}{\partial t}$

Momentum: $\sum\mathbf{F} = \frac{\partial}{\partial t}\int\mathbf{V}\rho d\forall + \sum(\dot{m}\mathbf{V})_{out} - \sum(\dot{m}\mathbf{V})_{in}$

Ang. Mom (Stationary):

$\sum\mathbf{M}_{ext} = \frac{\partial}{\partial t}\int_{CV}(\mathbf{r}\times\mathbf{V})\rho d\forall + \int_{CS}(\mathbf{r}\times\mathbf{V})\rho(\mathbf{V}\cdot\mathbf{n})dA$

Force of Jet on Vane:

$F_x = \dot{m}(V_{1,x} - V_{2,x}); \quad F_y = \dot{m}(V_{1,y} - V_{2,y})$

Energy: $\frac{P_1}{\gamma} + \alpha_1\frac{V_1^2}{2g} + z_1 + h_p = \frac{P_2}{\gamma} + \alpha_2\frac{V_2^2}{2g} + z_2 + h_t + h_L$

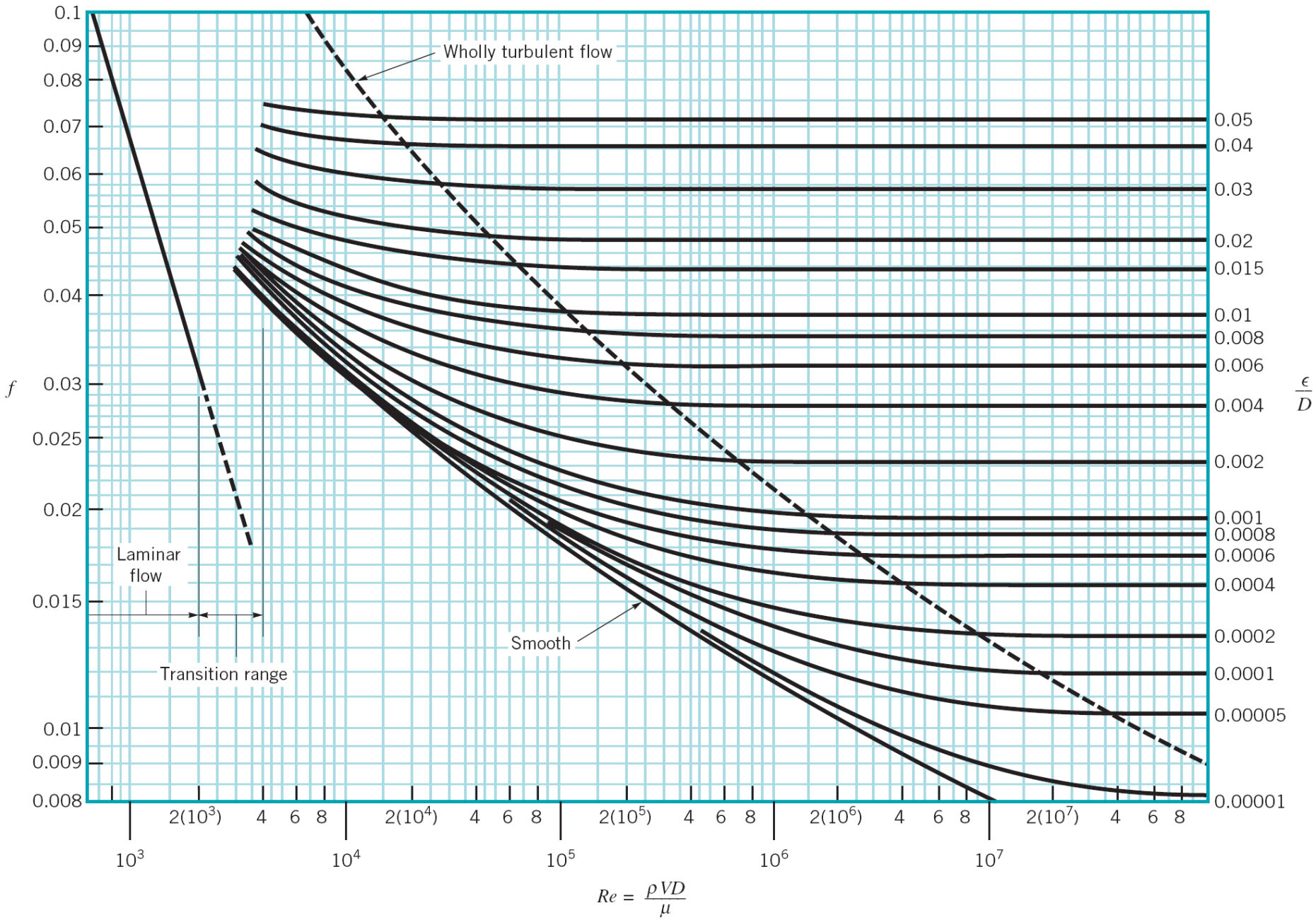
Factors: $\alpha = 2.0$ (Lam), $\alpha \approx 1.05$ (Turb)

Grade Lines: $EGL = HGL + \alpha V^2/2g$

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$$Re = 1.1 \times 10^5$$

$$\frac{\epsilon}{D} = 0.0006$$



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